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EE1030/EE2100: Matrix Theory

Lecture 24: Solving Systems of Linear Equations

Topics :

1. Solving Systems of Equations of the form $U\mathbf{x} = \mathbf{b}$
 2. How about $A\mathbf{x} = \mathbf{b}$?
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Solving Systems of Linear Equations

final goal: Solve a system of linear Equations of the form $A\vec{x} = \vec{b}$ where $A \in \mathbb{R}^{m \times n}$, $\vec{x} \in \mathbb{R}^n$ & $\vec{b} \in \mathbb{R}^m$

our approach: ① $A\vec{x} = \vec{b}$ when $m = n$ (no. of Eqn's = no. of unknowns)

② $A\vec{x} = \vec{b}$ when $m > n$ (overdetermined system of LE)

③ $A\vec{x} = \vec{b}$ when $m < n$ (underdetermined system of LE)

↓ Solution approaches: a) Gaussian elimination $\begin{matrix} \nearrow \text{FE} \\ \searrow \text{BS} \end{matrix}$

b) LU decomposition (few variants)

✓ c) QR decomposition (✓) → Robust

d) Iterative methods. → formulate an optimization Prob & sol to it, $A\vec{x} = \vec{b}$

QR decomposition approach: ① Express $A = QR$ where Q is an orthogonal matrix (or) (Pref.) orthonormal matrix.

$R \leftarrow$ upper triangular matrix.
② Solve System of Eqn's of the form $R\vec{x} = \vec{b}$

Solving Systems of Linear Equations

Consider system of LE of the form $U\vec{x} = \vec{b}$ where U is upper triangular & square
 $\vec{x} \in \mathbb{R}^n, \vec{b} \in \mathbb{R}^n$

$$\left. \begin{aligned} U_{11}x_1 + U_{12}x_2 + \dots + U_{1n}x_n &= b_1 \\ U_{22}x_2 + \dots + U_{2n}x_n &= b_2 \\ &\vdots \\ U_{nn}x_n &= b_n \end{aligned} \right\} \rightarrow \textcircled{I}$$

$$\leftrightarrow \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

from the last Eqn in the set $\textcircled{I} \Rightarrow x_n = b_n / U_{nn} \rightarrow \textcircled{1}$

from the second last Eqn $\Rightarrow U_{n-1,n-1}x_{n-1} + U_{n-1,n}x_n = b_{n-1}$

$$\Rightarrow x_{n-1} = \frac{b_{n-1} - U_{n-1,n}x_n}{U_{n-1,n-1}} \rightarrow \textcircled{2}$$

from the i^{th} Eqn in the set $\Rightarrow U_{ii}x_i + \sum_{j=i+1}^n U_{ij}x_j = b_i$

$$\Rightarrow x_i = \frac{b_i - \sum_{j=i+1}^n U_{ij}x_j}{U_{ii}} \rightarrow \textcircled{*}$$

Back Substitution

approach may fail if $U_{ii} = 0 \ \forall \ i=1, \dots, n-1 \Rightarrow$ does it indicate that Column/Row vectors of U are linearly dep?

Solving Systems of Linear Equations (ideas behind Gaussian elimination in brief) using $A \in \mathbb{R}^{3 \times 3}$

① Consider a system of linear equations

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \rightarrow \textcircled{1} \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \rightarrow \textcircled{2} \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \rightarrow \textcircled{3} \end{aligned} \right\} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

elementary row operations: $\left\{ \begin{aligned} R_2 &: a_{11}R_2 - a_{21}R_1 \\ R_3 &: a_{11}R_3 - a_{31}R_1 \end{aligned} \right\} \leftarrow$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b'_2 \\ b'_3 \end{bmatrix}$$

Next set of elementary row operations: $R_3: a'_{22}R_3 - a'_{32}R_2 =$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b'_2 \\ b''_3 \end{bmatrix}$$

System of LE of the form $U\vec{x} = \vec{b}$

Gaussian elimination a) Sequence of elementary row operations to

convert $A\vec{x} = \vec{b}$ to $U\vec{x} = \vec{b}'$ (forward elimination)

b) Back substitution

QR Factorization and Solving Systems of Linear Equations

